

CS 486/686

Value & Policy Iteration

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Lecture 14

RN 17.2 · PM 9.5.2–9.5.3



Learning goals

- Trace + implement the **value iteration** algorithm for solving an MDP.
- Trace + implement the **policy iteration** algorithm for solving an MDP.

Recap: $\pi^\star = \arg \max_a Q^\star$

From L13, the optimal policy is one greedy step from $V^\star$:

$$\pi^\star(s) = \arg \max_a Q^\star(s, a), \quad Q^\star(s, a) = R(s) + \gamma \sum_{s'} P(s' | s, a) V^\star(s').$$

Today's question

How do we actually *compute* $V^\star$?

Two algorithms: **value iteration** (iterate values) and **policy iteration** (iterate policies).

Value functions

- $V^\pi(s)$ — value of being in state s under policy π .
- $V^*(s)$ — value of s under the **optimal** policy π^* .
- $Q^\pi(s, a)$ — value of taking action a in s , then following π .
- $Q^*(s, a)$ — same, but under π^* .
- $\pi(a \mid s)$ — the policy, a distribution over actions per state.

"Value" = expected discounted future reward.

Formal definitions

Return (discounted future reward from time t):

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = R_{t+1} + \gamma G_{t+1}.$$

Value functions are expected returns:

$$V^\pi(s) = \mathbb{E}_\pi [G_t \mid s_t = s], \quad Q^\pi(s, a) = \mathbb{E}_\pi [G_t \mid s_t = s, a_t = a].$$

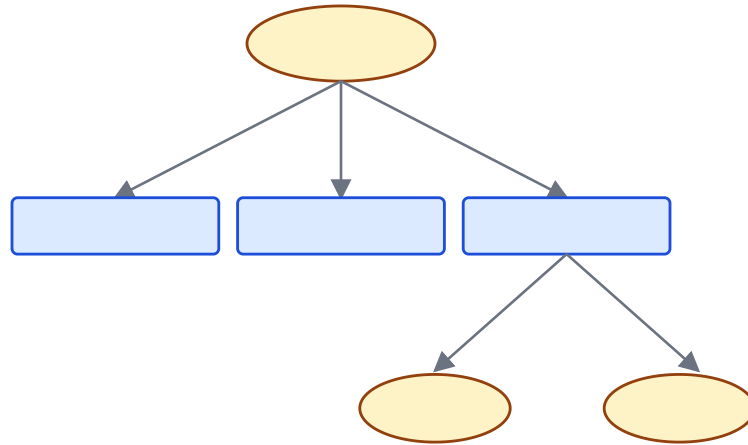
The two are related (each is a sum of the other):

$$V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a)$$

$$Q^\pi(s, a) = R(s) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s')$$

The backup diagram

A picture of the V/Q recursion: **policy** branches on actions; **environment** branches on next states.



The Bellman optimality equation

For the optimal policy, choose the *best* action at every state:

$$V^*(s) = \max_a Q^*(s, a), \quad Q^*(s, a) = R(s) + \gamma \sum_{s'} P(s' | s, a) V^*(s').$$

Substitute one into the other:

Bellman equation

$$V^*(s) = R(s) + \gamma \max_a \sum_{s'} P(s' | s, a) V^*(s').$$

V^* is the **unique** solution to this system of $|S|$ equations.

Apply Bellman to $V^*(s_{11})$

s_{11}	0.0	0.0	0.04
0.0	X	0.0	-1
0.0	0.0	0.0	+1

From s_{11} , each action gives a different mixture (0.8 intended + 0.1 each side):

$$V^*(s_{11}) = -0.04 + \gamma \max \left\{ \begin{array}{l} 0.8 V^*(s_{12}) + 0.1 V^*(s_{21}) + 0.1 V^*(s_{11}), \text{ // right} \\ 0.9 V^*(s_{11}) + 0.1 V^*(s_{12}), \text{ // up} \\ 0.9 V^*(s_{11}) + 0.1 V^*(s_{21}), \text{ // left} \\ 0.8 V^*(s_{21}) + 0.1 V^*(s_{12}) + 0.1 V^*(s_{11}) \text{ // down} \end{array} \right\}$$

Q1. Can we solve this system of Bellman equations in polynomial time?

No. The system is *nonlinear* because of the **max** — there is no general efficient solver. We'll use an **iterative** algorithm instead.

The value iteration algorithm

Treat the Bellman equation as an *update rule*. Let $V_i(s)$ be the i^{th} estimate.

value-iteration(MDP, ϵ)

1. Initialize $V_0(s)$ arbitrarily (e.g. 0).
2. For $i = 0, 1, 2, \dots$: for each state s , apply the Bellman update

$$V_{i+1}(s) \leftarrow R(s) + \gamma \max_a \sum_{s'} P(s' \mid s, a) V_i(s').$$

3. Stop when $\max_s |V_{i+1}(s) - V_i(s)| < \epsilon$.

Guarantee: $V_i \rightarrow V^*$ as $i \rightarrow \infty$.

- **Synchronous:** keep V_i frozen while computing all of V_{i+1} .
- **Asynchronous:** update entries in place, in any order. Often converges faster.

Apply VI to the grid

Setup: $\gamma = 1$, $R(s) = -0.04$ for non-goal states. Terminals: $V(s_{24}) = -1$, $V(s_{34}) = +1$. Init $V_0(s) = 0$ for all non-terminals.

$V_0(s)$

0	0	0	0
0	x	0	-1
0	0	0	+1

After one Bellman update, only the cells adjacent to a goal will see something different from -0.04 .

One iteration: $V_1(s)$

Apply $V_1(s) \leftarrow R(s) + \gamma \max_a \sum_{s'} P(s' | s, a) V_0(s')$ at every cell. Two cells worth a closer look:

$V_1(s)$

-0.04	-0.04	-0.04	-0.04
-0.04	X	-0.04	-1
-0.04	-0.04	0.76	+1

Q2. $V_1(s_{23})$ (yellow): all neighbors are 0; best action is Left (avoids -1). $\Rightarrow -0.04 + 0 = \boxed{-0.04}$.

Q3. $V_1(s_{33})$ (green): Right has $0.8 \cdot 1 + 0.1 \cdot 0 + 0.1 \cdot 0 = 0.8$. $\Rightarrow -0.04 + 0.8 = \boxed{0.76}$.

The +1 reward has "leaked" one cell up; the -1 hasn't propagated because we avoid it.

Two iterations: $V_2(s)$

Apply Bellman again, using V_1 as input. Three cells worth a closer look:

$V_2(s)$

-0.08	-0.08	-0.08	-0.08
-0.08	X	0.464	-1
-0.08	0.56	0.832	+1

Q4. $V_2(s_{33})$: Right gives $0.8(+1) + 0.1(-0.04) + 0.1(0.76) = 0.872. \Rightarrow$
 $-0.04 + 0.872 = \boxed{0.832}.$

Q5. $V_2(s_{23})$: Down gives $0.8(0.76) + 0.1(-0.04) + 0.1(-1) = 0.504. \Rightarrow$
 $-0.04 + 0.504 = \boxed{0.464}.$

Q6. $V_2(s_{32})$: Right gives $0.8(0.76) + 0.1(-0.04) + 0.1(-0.04) = 0.6. \Rightarrow$
 $-0.04 + 0.6 = \boxed{0.56}.$

Values propagate one cell per iteration. After enough sweeps they converge to the L13 V^* .

Policy iteration

Insight: if one action is clearly better, you don't need precise utility values — just enough to rank actions.

policy-iteration(MDP)

1. Start from an arbitrary policy π_0 .
2. Repeat:
 - **Policy evaluation:** compute $V^{\pi_i}(s)$, the utility of every state under π_i .
 - **Policy improvement:** $\pi_{i+1}(s) = \arg \max_a \sum_{s'} P(s' \mid s, a) V^{\pi_i}(s')$.
3. Until $\pi_{i+1} = \pi_i$.

Terminates in finitely many steps: there are finitely many policies and each iteration strictly improves.

Policy evaluation vs Bellman: the max disappears

In PE the action is *fixed* by π , so the system becomes linear in V :

Policy evaluation (linear)

$$V^\pi(s) = R(s) + \gamma \sum_{s'} P(s' | s, \pi(s)) V^\pi(s')$$

No **max** — the action is fixed. $|S|$
linear equations in $|S|$ unknowns.

linear: easy to solve

Bellman optimality (nonlinear)

$$V^*(s) = R(s) + \gamma \max_a \sum_{s'} P(s' | s, a) V^*(s')$$

The **max** makes it nonlinear — need
an iterative algorithm (VI).

nonlinear: harder

Policy improvement reintroduces the **arg max**: $\pi_{i+1}(s) = \arg \max_a \sum_{s'} P(s' | s, a) V^{\pi_i}(s')$.

How do we perform policy evaluation?

Two ways to solve the linear system $V^\pi(s) = R(s) + \gamma \sum_{s'} P(s' \mid s, \pi(s)) V^\pi(s')$:

Exactly

Linear algebra: a single solve of an $|S| \times |S|$ system.

cost: $O(|S|^3)$

Iteratively (cheap)

Run m simplified Bellman updates (no max):

$$V_{j+1}(s) \leftarrow R(s) + \gamma \sum_{s'} P(s' \mid s, \pi(s)) V_j(s')$$

cost: $O(m \cdot |S| \cdot |A_{\text{avg}}|)$

In practice, a small m is usually enough — PI converges in far fewer outer iterations than VI.

Learning goals (recap)

- ✓ Trace and implement **value iteration** for solving an MDP.
- ✓ Trace and implement **policy iteration** for solving an MDP.

Next: reinforcement learning

Today we knew the dynamics P and rewards R of the MDP. What if we *don't*?

L15: learn the policy by **interacting with the environment**.